

TADD MAO

Mathematics & Computer Science, University of Toronto

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RESEARCH INTERESTS

AI verification: how can we know, with formal certainty that an AI system did exactly what we asked, and how can we build a systems that can be trusted rather than trusting the model directly?

EDUCATION

University of Toronto

Sep. 2023 – Apr. 2027 (expected)

H.B.Sc., Computer Science Major (Focus in Artificial Intelligence) & Mathematics Major

Toronto, ON

- Cumulative GPA: **3.99 / 4.00**.
- Selected grades: Mathematical Logic (**100%**), Theory of Computation (99%), Abstract Mathematics (97%), Group Theory (97%), Real Analysis (94%), Advanced Calculus (94%), Machine Learning (90%).
- Additional coursework: Deep Learning, Artificial Intelligence, Natural Language Processing, Computational Linguistics, Topology, Graph Theory, Programming Languages.

PAPERS

Does the Proof Prove It That Way? Faithful Formalization of Euclid's Elements Proofs

Tadd Mao, Tianjun Zhong, Dhruva Arekar, Yuming Feng, One An, Jiani Huang, Xujie Si, Ziyang Li.

Preprint; under review at AAAI-27. [arXiv:2608.15432](https://arxiv.org/abs/2608.15432)

- Defined five necessary conditions for how faithfully a formal Lean proof reflects the natural-language proof it claims to represent, and built PISTIS, an oracle-guided proof search that generates proofs meeting them.
- On Euclid's *Elements* (Books I–III, 92 propositions): compiles at least 33× faster than prior work and is preferred 2.89× (127 blinded human reviews) and 5.2× (LLM judge) as often over the prior state of the art.
- Surfaced 52 gaps in Euclid's proofs and 2 genuine citation errors in the translated text.

RESEARCH EXPERIENCE

Faithfulness of autoformalized mathematics

June 2026 – present

Johns Hopkins (Ziyang Li) & U. Pennsylvania; with Xujie Si (Toronto)

Collaboration

- First-authored the AAAI-27 submission above (Does the Proof prove it That Way?...)
- Ongoing: a theory-level *faithfulness metric* for AI-generated formalizations of natural-language mathematics (in preparation, targeting ICLR 2027).

Autoformalization in Euclidean Geometry (NSERC USRA)

June 2026 – present

University of Toronto, advised by Prof. Xujie Si

Toronto, ON

- Building the first complete formalization of Euclid's *Elements* in Lean, extending Avigad et al.'s SYSTEM E with a new axiom set.
- Developing methods for evaluating the quality of a given set of axioms.
- Early work on formalizing common-sense reasoning.

Multi-agent reinforcement learning

Aug. 2025 – present

University of Toronto, advised by Prof. Peter Marbach

Toronto, ON

- Studying when *centralized* (agents together learn one joint value function) versus *decentralized* (each agent learns its own value function) is preferable in which systems, where the system is characterized by parameters rather than on a case by case basis for more generality.
- Aimed at AAMAS 2027

Laser Scanning Research Assistant

University of Northern British Columbia, under Dr. Wenbo Zheng

Summer 2022

Prince George, BC

- Built the lab's first laser-scanning accuracy metric (custom MATLAB); authored a **workflow guide** that accelerated new-assistant onboarding.

OPEN-SOURCE CONTRIBUTION

- **Mathlib (Lean 4)** — authored and merged a lemma (`Int.closure_eq_zmultiples`) into MATHLIB, the community's core formal-mathematics library; wrote the proof independently and refined it through community review and design discussion. [PR #39435](#)

TEACHING

Teaching Assistant — MAT309 (Mathematical Logic)

University of Toronto

Fall 2026

Toronto, ON

- Lead 2 of the 3 weekly tutorial sections; also marking and invigilation.

PROFESSIONAL SERVICE

- Sub-reviewer, AAAI 2027.

HONORS & AWARDS

- NSERC Undergraduate Student Research Award (2026).
- Dean's List Scholar (2024, 2025, 2026 — all completed years).
- Innis College Exceptional Achievement Award (2024).
- Later Life Learning Scholarship (2025).

SKILLS

Formal methods: Lean 4, Mathlib

Programming: Python, C/C++, Java, R, MATLAB

ML & tools: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas; Slurm, Linux, Git

Natural languages: English, French, Mandarin

PERSONAL

- Winner, Prince George Symphony Orchestra Young Artists Competition — performed the first movement of Rachmaninoff's Piano Concerto No. 2 as piano soloist with the PGSO ([video](#))
- Toured with professional musicians (Yu Yu Liu, Simon Macdonald, Roxanne Dykstra), performing violin in ensemble and piano solo, including a standing ovation for Liszt's *La Campanella*.
- Royal Conservatory of Music: ARCT diploma (piano); Level 10 (violin).

